

Event Booking Management System (EBMS)

Project Documentation

1. Overview

EBMS is a web platform that connects **event organizers** with **ticket-buying attendees**. Organizers create and publish events and sell tickets directly through the platform; attendees browse events and purchase tickets online, with payments processed securely through **Stripe**. The platform acts as the marketplace between the two sides, handling bookings, payments, ticketing, and communication in one place.

What it solves:

- Eliminates overbooking with real-time capacity tracking
- Centralizes attendee and booking records
- Automates booking confirmations, reminders, and payment notifications
- Gives organizers tools to manage events and track revenue
- Gives platform staff full administrative control and audit history

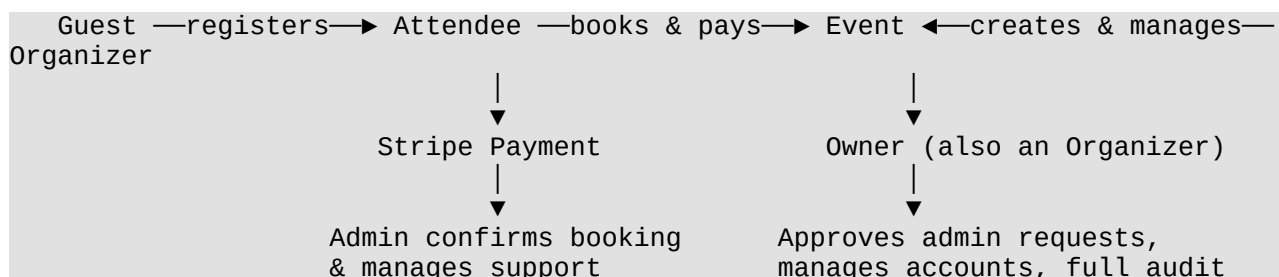
What it delivers:

- Automated ticket booking and payment
 - Organizer event management and analytics
 - Administrative oversight with full audit logs
 - Automated notifications across every step of the booking lifecycle
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2. Who Uses the Platform

Role	What they do
Guest	Browses events without an account; can send a general inquiry; must register to book
Attendee	Registered user who browses events, books tickets, manages bookings, downloads QR tickets
Organizer	Creates and publishes events, sets ticket categories and pricing, tracks revenue and attendance
Admin	Runs daily platform operations — confirms bookings, manages reports, handles support inbox

Role	What they do
Owner	Platform owner; approves administrative changes, manages staff accounts, holds full audit access



3. How a Booking Works

1. Attendee chooses an event, ticket category, and quantity
2. The system reserves the tickets for 10 minutes while checkout is completed
3. Attendee pays securely through **Stripe**
4. On successful payment, the booking is confirmed and a QR-code ticket is issued
5. A confirmation notification is sent immediately
6. Admin staff can review pending bookings, and mark attendance once the event begins

Attendees can cancel or reschedule a booking up until the event's refund deadline. Organizers and admins receive notifications throughout, and the system keeps both sides informed automatically at every step.

4. Platform Features

For Attendees

- Browse and search events, view full event details
- Book tickets, pay securely via Stripe, receive QR-code tickets
- View booking history and status, cancel or reschedule bookings
- Manage profile, receive automated notifications and reminders
- Contact support (general questions, complaints, or suggestions)

For Organizers

- Create, publish, update, and cancel events
- Configure ticket categories, pricing, and discounts

- Set refund and ticket-transfer policies
- Track bookings, revenue, and attendance in real time
- Export attendee lists and generate event reports

For Admin Staff

- Confirm or reject booking requests
- Search and manage attendee and booking records
- Mark attendance at events
- Generate daily/weekly reports, revenue summaries, and no-show reports
- Manage the support inbox for general inquiries
- Submit requests for platform changes (services, FAQs, offers) for owner approval

For the Owner

- Approve or reject all administrative change requests
- Create and manage Admin and Organizer accounts
- Full visibility into all system activity (complete audit log)
- Handle complaints and suggestions
- Manually activate or deactivate promotional offers and discounts

Platform-wide

- Automated email notifications for every booking, payment, and account event
 - Secure payment processing — no card details are ever stored on the platform
 - Mobile-responsive design across every page
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5. Reliability & Security

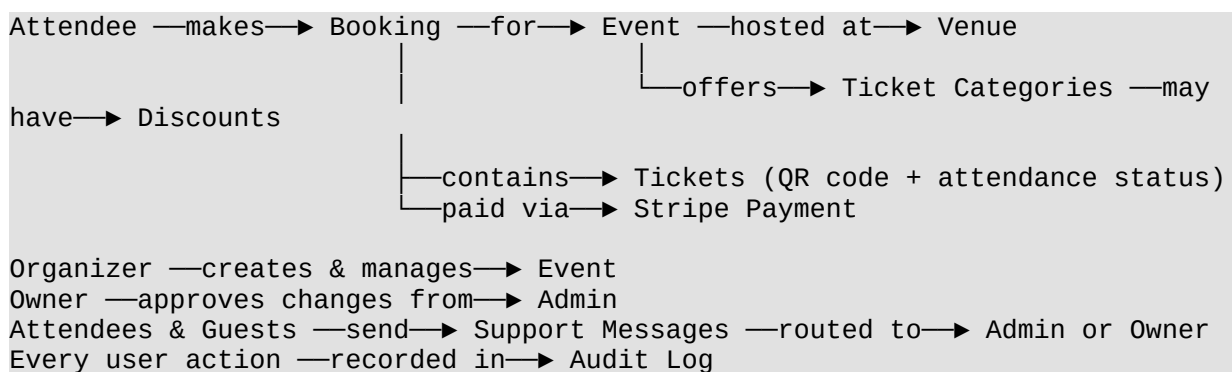
- Built to handle multiple concurrent users, including traffic spikes during high-demand ticket releases
 - All protected actions require authentication; sensitive data is encrypted in transit and at rest
 - No payment card data is stored on the platform — Stripe handles all card information
 - Booking and payment operations are transactionally consistent, preventing double-booking even under simultaneous requests
 - Every critical action is logged with a timestamp and user ID; logs are retained for a minimum of two years for audit and compliance
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6. Platform Rules

- A booking is only confirmed after successful payment
 - Reserved tickets are released automatically if checkout isn't completed within 10 minutes
 - Cancellations are accepted up to 24 hours before the event start; tickets are non-transferable unless the organizer permits it
 - An event must have at least one active ticket category before it can be published
 - Deactivated Admin or Organizer accounts cannot log in
 - New staff accounts require a password change on first login
 - Attendance can only be marked once the event has started
 - Changes to platform services, FAQs, or promotional offers require Owner approval before taking effect
 - Only the Owner can manually activate or deactivate discounts
 - Guests can browse and send general inquiries only; booking requires registration
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7. Information Managed by the System

The platform tracks everything needed to run bookings end-to-end: user accounts, events and venues, ticket categories and pricing, discounts, bookings and individual tickets, payments, notifications, support messages, and a complete activity log.



8. Platform Pages

The platform is fully responsive across desktop and mobile, with a clean, modern interface (blue as the primary brand color) and consistent card-based layouts throughout.

Public Pages | Page | Contents | |---|---| | **Home** | Search bar, featured events, platform statistics, call-to-action | | **Events Listing** | Filters by category, location, date, and price; sorting; grid/list view; search | | **Event Details** | Banner image, full event info, ticket selection with quantity, price breakdown, booking button, related events | | **Login / Register** | Account access and creation, password recovery, social sign-in options |

Booking | Page | Contents | |---|---| | **Checkout** | Order summary, attendee details form, secure Stripe payment, final price breakdown, confirmation |

Attendee Dashboard | Page | Contents | |---|---| | **My Bookings** | Upcoming bookings, QR-code tickets, booking status, cancel option | | **Profile** | Personal details management | | **Notifications** | Booking and event updates |

Organizer Dashboard | Page | Contents | |---|---| | **Event Management** | Create, publish, edit, and cancel events; configure ticket categories and pricing | | **Revenue & Bookings** | Real-time revenue tracking, attendee export, attendance overview |

Admin / Owner Dashboard | Page | Contents | |---|---| | **Overview** | Key stats at a glance | | **Event & Booking Management** | Full event and booking tables with confirm/reject/cancel actions | | **User Management** | Manage Organizer and Admin accounts | | **Reports** | Revenue summaries, daily/weekly reports, no-show reports, export | | **Approvals** (*Owner only*) | Review and approve/reject administrative change requests | | **Audit Log** (*Owner only*) | Full history of platform activity |

9. Technology

- **Backend:** ASP.NET Core MVC, organized in clean, layered architecture for maintainability
 - **Database:** MySQL with Entity Framework Core
 - **Payments:** Stripe for secure checkout, transactions, and refunds
 - **Frontend:** Responsive, mobile-friendly interface with a consistent design system
 - **Notifications:** Automated email notifications triggered by booking, payment, and account events
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10. What This Means for You

EBMS gives you a single, automated platform to launch events, sell tickets, and manage attendees — without the overbooking errors, manual tracking, and missed communications of spreadsheet-based or partially digital systems. Organizers get full visibility into their events and revenue, attendees get a smooth, secure booking experience, and platform staff retain complete oversight and control.